

```
import os
import pandas as pd
os.chdir(r"C:\Users\cebal\OneDrive\Documentos\Grandes Bases de Datos I")
```

```
estudiantes = pd.read_csv('EjemploEstudiantes.csv', sep=";", decimal=",") #cargando los datos de los estudiante
estudiantes
```

```
Unnamed: 0  Matematicas  Ciencias  Espanol  Historia  EdFisica
0      Lucia          7.0        6.5      9.2        8.6        8.0
1      Pedro          7.5        9.4      7.3        7.0        7.0
2       Ines          7.6        9.2      8.0        8.0        7.5
3       Luis          5.0        6.5      6.5        7.0        9.0
4      Andres          6.0        6.0      7.8        8.9        7.3
5        Ana          7.8        9.6      7.7        8.0        6.5
6      Carlos          6.3        6.4      8.2        9.0        7.2
7       Jose          7.9        9.7      7.5        8.0        6.0
8      Sonia          6.0        6.0      6.5        5.5        8.7
9      Maria          6.8        7.2      8.7        9.0        7.0
```

```
etiquetas = estudiantes.iloc[:, 0]
etiquetas
```

```
0      Lucia
1      Pedro
2       Ines
3       Luis
4      Andres
5        Ana
6      Carlos
7       Jose
8      Sonia
9      Maria
Name: Unnamed: 0, dtype: str
```

```
from numpy import array
from sklearn.decomposition import TruncatedSVD

A = estudiantes
# Remover columna no numérica
A = A.drop(estudiantes.columns[0], axis = 1)
A
```

```
Matematicas  Ciencias  Espanol  Historia  EdFisica
0          7.0        6.5      9.2        8.6        8.0
1          7.5        9.4      7.3        7.0        7.0
2          7.6        9.2      8.0        8.0        7.5
3          5.0        6.5      6.5        7.0        9.0
4          6.0        6.0      7.8        8.9        7.3
5          7.8        9.6      7.7        8.0        6.5
6          6.3        6.4      8.2        9.0        7.2
7          7.9        9.7      7.5        8.0        6.0
8          6.0        6.0      6.5        5.5        8.7
9          6.8        7.2      8.7        9.0        7.0
```

```
from numpy import diag
from numpy import zeros
from scipy.linalg import svd
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
U, s, VT = svd(A) # Obtención de matrices cuyo producto arroja la matriz A
sigma = zeros([A.shape[0], A.shape[1]]) # Creación de matriz de ceros de la misma dimensión que A
sigma[:A.shape[1], :A.shape[1]] = diag(s)
```

```
print("Matriz U:")
print(U)
```

```
Matriz U:
[[-3.30904778e-01 -2.98442471e-01  1.95957496e-01 -5.49129984e-01
  3.41498230e-01  3.14150349e-02 -1.23395489e-01  1.44668847e-01
 -3.94701586e-01 -3.89792548e-01]
 [-3.20798129e-01  3.55909651e-01 -2.40201517e-01 -6.92729616e-02
  2.04368756e-01 -4.82498850e-01  2.12566116e-01 -5.29979923e-01
 -2.91960519e-01  1.48477689e-01]
 [-3.38871518e-01  2.14752307e-01 -1.01355072e-01  1.12970083e-02
  1.76844101e-01 -1.27934010e-01 -3.25968835e-01 -5.72499990e-02
  7.00049221e-01 -4.25915443e-01]
 [-2.86319969e-01 -3.47796853e-01 -4.56775340e-01  6.26123724e-01
  3.29136285e-01  7.19838459e-02 -5.36517108e-04  2.16809671e-01
 -1.85075729e-01 -4.01432467e-02]
 [-3.03747698e-01 -3.04664300e-01  2.89885414e-01  2.58403657e-01
 -5.09400304e-01 -1.58301296e-01 -4.86485858e-01 -3.39970765e-01
 -1.65040306e-01  8.72429330e-03]
 [-3.33011382e-01  3.96082489e-01  2.51606890e-02  5.94976208e-02
 -5.78210369e-02  8.06621727e-01  1.34546582e-02 -2.29782306e-01
 -1.43611096e-01 -2.14962316e-02]
 [-3.13003741e-01 -2.44042780e-01  3.27809809e-01  1.29815059e-01
 -1.75652482e-01 -3.52533633e-02  7.64914176e-01 -5.44057025e-02
  2.27556312e-01 -2.15007938e-01]
 [-3.28787999e-01  4.75531970e-01  9.14254492e-02  7.72474797e-02
 -3.06415596e-01 -2.40842651e-01 -5.18130571e-03  6.78727567e-01
 -1.96390068e-01  1.67121472e-02]
 [-2.73896980e-01 -2.64522156e-01 -6.03090161e-01 -4.52523149e-01
 -4.34056879e-01  9.18463660e-02  6.22904133e-02  6.57998448e-02
  1.69210215e-01  2.29849057e-01]
 [-3.26383007e-01 -1.08710765e-01  3.47541493e-01 -5.99185845e-02
  3.55001709e-01  3.57424428e-02 -8.65047599e-02  1.07552174e-01
  2.60388740e-01  7.36969606e-01]]
```

```
print("Matriz Sigma")
print(sigma)
```

```
Matriz Sigma
[[53.21335049  0.          0.          0.          0.          ]
 [ 0.          5.35631448  0.          0.          0.          ]
 [ 0.          0.          3.80560674  0.          0.          ]
 [ 0.          0.          0.          1.46904724  0.          ]
 [ 0.          0.          0.          0.          0.47799818]
 [ 0.          0.          0.          0.          0.          ]
 [ 0.          0.          0.          0.          0.          ]
 [ 0.          0.          0.          0.          0.          ]
 [ 0.          0.          0.          0.          0.          ]
 [ 0.          0.          0.          0.          0.          ]]
```

```
print("Matriz VT")
print(VT)
```

```
Matriz VT
[[-0.40556463 -0.45759727 -0.46123811 -0.47103299 -0.43761787]
 [ 0.30388113  0.70497323 -0.20845983 -0.2165757  -0.56595801]
 [-0.00426062 -0.27229676  0.31573346  0.5963283  -0.6859601 ]
 [-0.56293162  0.34606734 -0.50555254  0.5444058  0.1066974 ]
 [-0.65288853  0.31576789  0.62332408 -0.28149203 -0.07909827]]
```

```
# Para reconstruir la matriz original
B = U.dot(sigma.dot(VT))
print(B)
```

```
[[7.  6.5 9.2 8.6 8. ]
 [7.5 9.4 7.3 7.  7. ]
 [7.6 9.2 8.  8.  7.5]
 [5.  6.5 6.5 7.  9. ]
 [6.  6.  7.8 8.9 7.3]
 [7.8 9.6 7.7 8.  6.5]
 [6.3 6.4 8.2 9.  7.2]
 [7.9 9.7 7.5 8.  6. ]
 [6.  6.  6.5 5.5 8.7]
 [6.8 7.2 8.7 9.  7. ]]
```

```
# Reducción a Dos Dimensiones
n_elements = 2
```

```
UReduced2 = U[:, :n_elements]
print("U Reducida:")
print(UReduced2)
```

```
U Reducida:
[[-0.33090478 -0.29844247]
 [-0.32079813  0.35590965]
 [-0.33887152  0.21475231]
 [-0.28631997 -0.34779685]
 [-0.3037477  -0.3046643 ]
 [-0.33301138  0.39608249]
 [-0.31300374 -0.24404278]
 [-0.328788    0.47553197]
 [-0.27389698 -0.26452216]
 [-0.32638301 -0.10871076]]
```

```
SigmaReduced2 = sigma[:n_elements, :n_elements]

print("Matriz Sigma Reducida")
print(SigmaReduced2)
```

```
Matriz Sigma Reducida
[[53.21335049  0.          ]
 [ 0.          5.35631448]]
```

```
VTReduced2 = VT[:n_elements, :]

print("VT Reducida")
print(VTReduced2)
```

```
VT Reducida
[[-0.40556463 -0.45759727 -0.46123811 -0.47103299 -0.43761787]
 [ 0.30388113  0.70497323 -0.20845983 -0.2165757  -0.56595801]]
```

```
AReduced2 = UReduced2.dot(SigmaReduced2.dot(VTReduced2))
print("Matriz A transformada:")
print(AReduced2)
```

```
Matriz A transformada:
[[6.65563617 6.93068907 8.45496902 8.64041633 8.61053016]
 [7.50259774 9.15546106 7.47627701 7.62801112 6.39154032]
 [7.66288835 9.06253486 8.0774837  8.24477427 7.24032869]
 [5.61309793 5.65867129 7.41578628 7.58014066 7.72189396]
 [6.05941975 6.24591247 7.7953721  7.96693512 7.99698132]
 [7.83156613 9.60455223 7.73118336 7.88753689 6.55416982]
 [6.35785122 6.70021013 7.95486409 8.12861624 8.02875678]
 [7.86973795 9.80171752 7.53881312 7.68951159 6.21497236]
 [5.48053658 5.67061811 7.01789357 7.17215282 7.18015541]
 [6.86687283 7.53702066 8.13213661 8.30697925 7.93006916]]
```

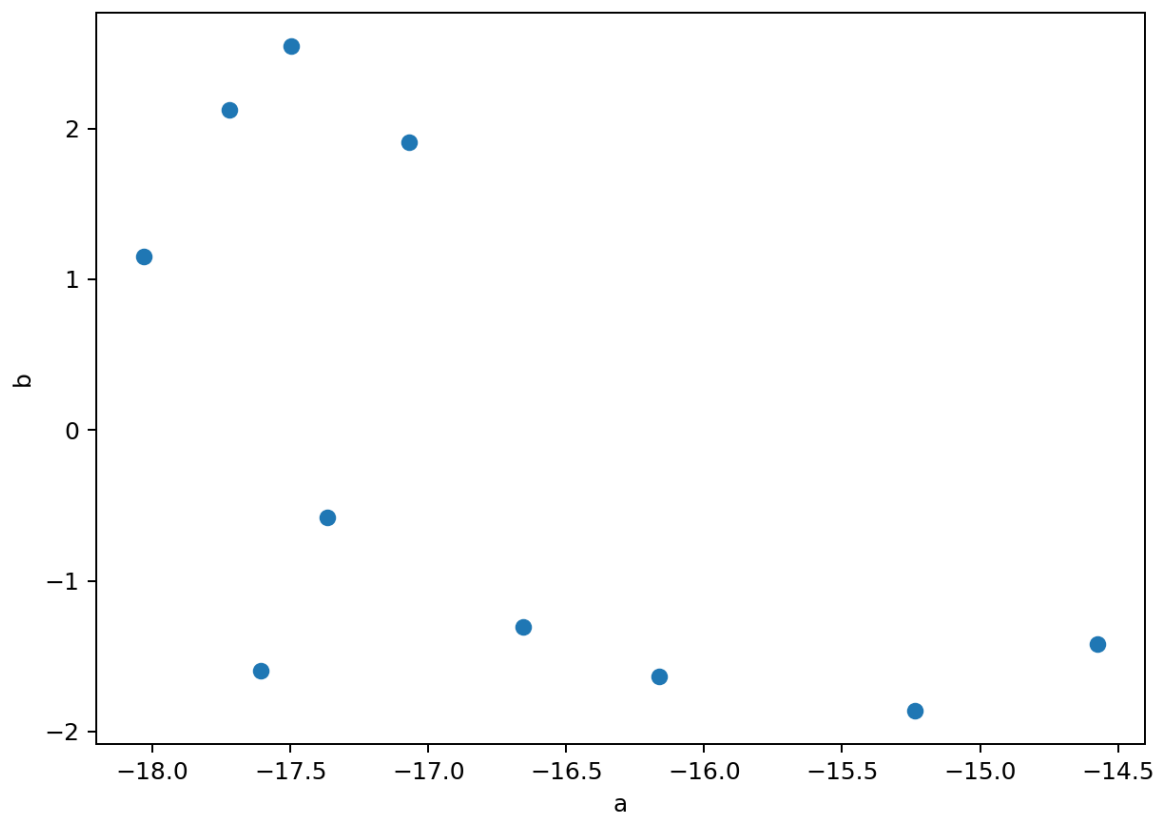
```
T2 = UReduced2.dot(SigmaReduced2)

print("Matriz T (2 Dimensiones):")
print(T2)
```

```
Matriz T (2 Dimensiones):
[[-17.60855193 -1.59855173]
 [-17.07074326  1.90636402]
 [-18.03248887  1.15028089]
 [-15.23604486 -1.86290932]
 [-16.1634327  -1.6318778 ]
 [-17.7206514  2.12154237]
 [-16.65597775 -1.30716988]
 [-17.49591103  2.54709878]
 [-14.57497601 -1.41686385]
 [-17.36793336 -0.58228904]]
```

```
df = pd.DataFrame(T2, columns=["a", "b"])
df.plot(kind="scatter", x="a", y="b")
plt.show()
```

<Figure size 640x480 with 1 Axes>



```
df = pd.DataFrame(T2, columns=["a", "b"])

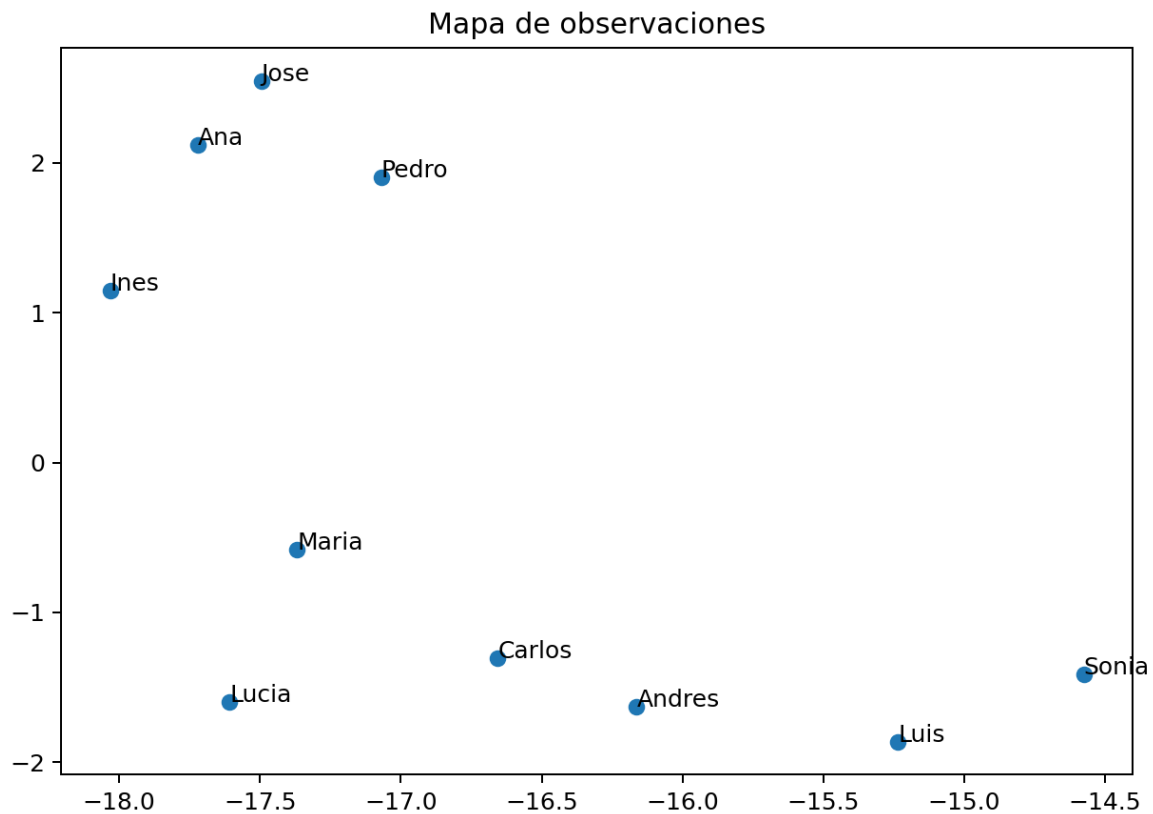
x = df.iloc[:, 0]
y = df.iloc[:, 1]

x = x.to_numpy()
y = y.to_numpy()

fig, ax = plt.subplots()
ax.set_title("Mapa de observaciones")
ax.scatter(x,y)

for i, txt in enumerate(etiquetas):
    ax.annotate(txt, (x[i], y[i]))
```

<Figure size 640x480 with 1 Axes>



```
# Reducción a 3 Dimensiones
n_elements = 3

UReduced3 = U[:, :n_elements]

SigmaReduced3 = sigma[:n_elements, :n_elements]

VTReduced3 = VT[:n_elements, :]

AReduced3 = UReduced3.dot(SigmaReduced3.dot(VTReduced3))

print("Matriz A transformada:")
print(AReduced3)
```

```
Matriz A transformada:
[[6.65245886 6.72762726 8.69042319 9.08512051 8.09898421]
 [7.50649243 9.40437093 7.1876611 7.08289996 7.01858503]
 [7.66453175 9.1675645 7.95569976 8.01475998 7.50491554]
 [5.6205042 6.13200674 6.8669445 6.54353881 8.91430342]
 [6.05471948 5.94551744 8.14368606 8.62479847 7.24023707]
 [7.83115817 9.57847935 7.76141538 7.94463634 6.48848799]
 [6.35253602 6.36051578 8.34874639 8.87254488 7.17301112]
 [7.86825556 9.7069775 7.64866605 7.89699168 5.97630674]
 [5.49031524 6.29557293 6.29324613 5.80350542 8.75451889]
 [6.86123771 7.17687926 8.54972766 9.09568679 7.02281405]]
```

```
T3 = UReduced3.dot(SigmaReduced3)

print("Matriz T (3 Dimensiones):")
print(T3)
```

```
Matriz T (3 Dimensiones):
[[-17.60855193 -1.59855173 0.74573717]
 [-17.07074326 1.90636402 -0.91411251]
 [-18.03248887 1.15028089 -0.38571754]
 [-15.23604486 -1.86290932 -1.73830731]]
```

```

[-16.1634327 -1.6318778  1.10318989]
[-17.7206514  2.12154237  0.09575169]
[-16.65597775 -1.30716988  1.24751522]
[-17.49591103  2.54709878  0.34792931]
[-14.57497601 -1.41686385 -2.29512399]
[-17.36793336 -0.58228904  1.32260625]]

```

```

from mpl_toolkits import mplot3d
%matplotlib inline

```

```

df3 = pd.DataFrame(T3, columns=["a", "b", "c"])
df3

```

```

a         b         c
0 -17.608552 -1.598552  0.745737
1 -17.070743  1.906364 -0.914113
2 -18.032489  1.150281 -0.385718
3 -15.236045 -1.862909 -1.738307
4 -16.163433 -1.631878  1.103190
5 -17.720651  2.121542  0.095752
6 -16.655978 -1.307170  1.247515
7 -17.495911  2.547099  0.347929
8 -14.574976 -1.416864 -2.295124
9 -17.367933 -0.582289  1.322606

```

```

from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure()
ax3d = fig.add_subplot(111, projection="3d")

arrayx = np.array(df3["a"])
arrayy = np.array(df3["b"])
arrayz = np.array(df3["c"])

ax3d.scatter(arrayx, arrayy, arrayz)

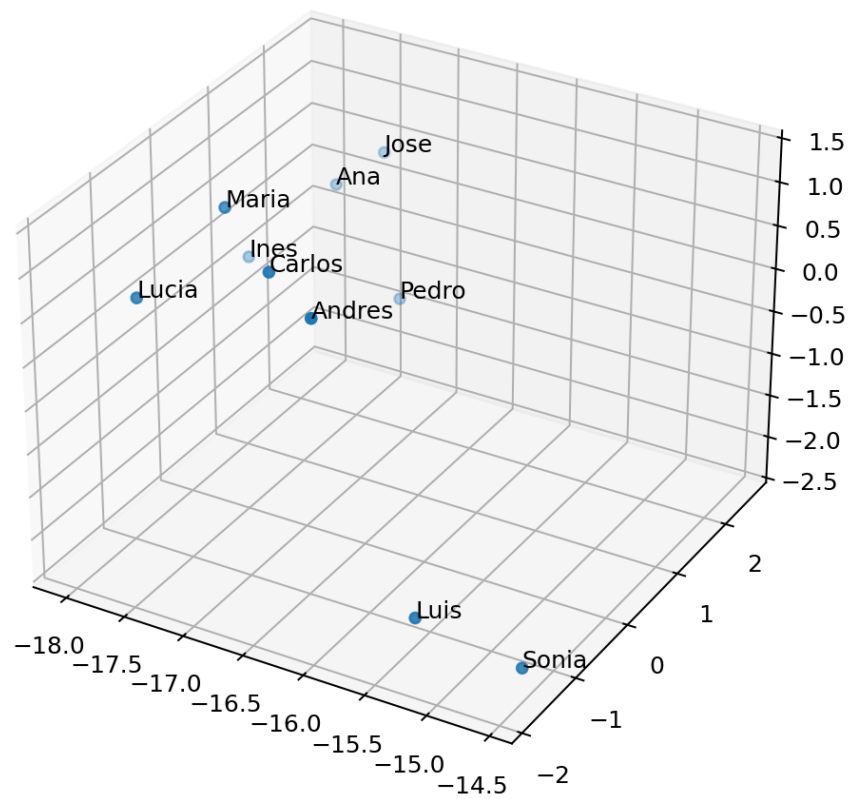
for x, y, z, label in zip(arrayx, arrayy, arrayz, etiquetas):
    ax3d.text(x, y, z, label)

plt.title("Datos en 3D")
plt.show()

```

<Figure size 640x480 with 1 Axes>

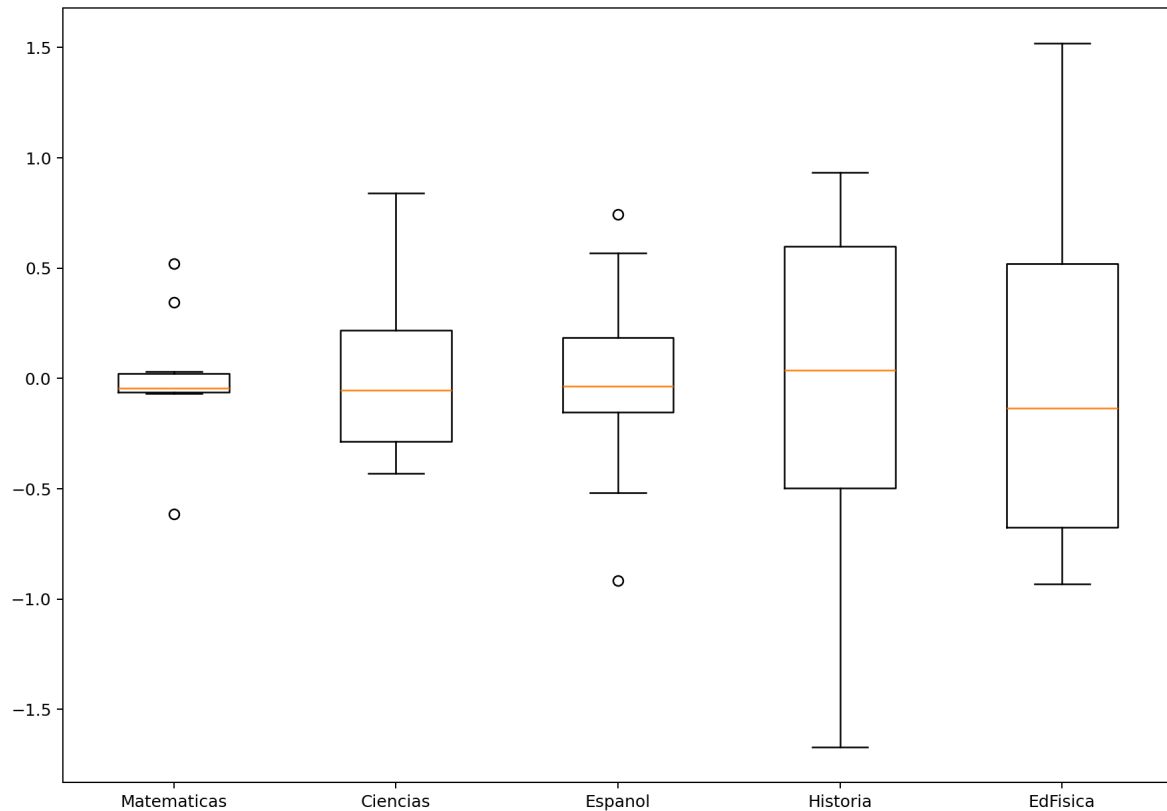
Datos en 3D



```
plt.figure(figsize=(10,7))

Error2 = B - AReduced2
dfError2 = pd.DataFrame(Error2, columns=["Matematicas", "Ciencias", "Espanol", "Historia", "EdFisica"])
dfError2.boxplot()
```

<Axes: >
<Figure size 1000x700 with 1 Axes>



```
print("Errores promedio por columna")
Error2.mean(axis=0)
```

Errores promedio por columna

```
array([-2.04650684e-05,  1.32612606e-02, -1.94778869e-02, -2.45074289e-02,
        3.30602005e-02])
```

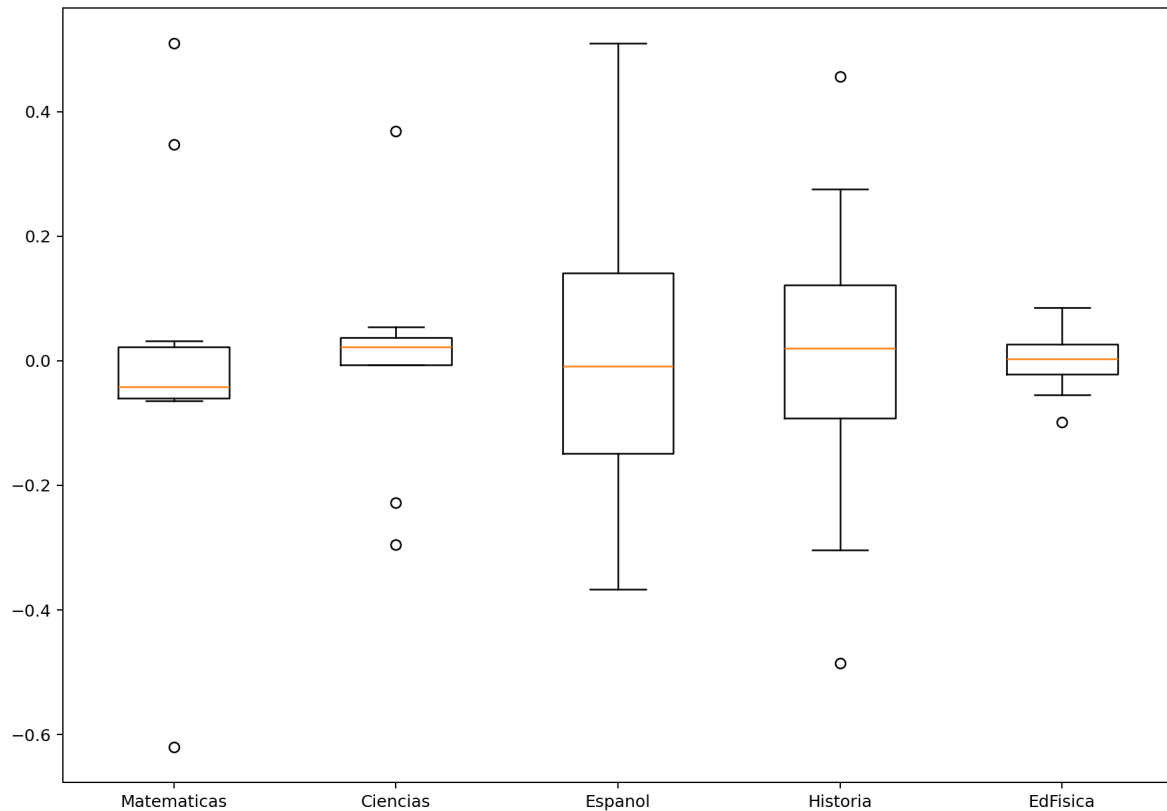
```
Error3 = B - AReduced3
print(Error3)
```

```
[[ 0.34754114 -0.22762726  0.50957681 -0.48512051 -0.09898421]
 [-0.00649243 -0.00437093  0.1123389  -0.08289996 -0.01858503]
 [-0.06453175  0.0324355   0.04430024 -0.01475998 -0.00491554]
 [-0.6205042   0.36799326 -0.3669445   0.45646119  0.08569658]
 [-0.05471948  0.05448256 -0.34368606  0.27520153  0.05976293]
 [-0.03115817  0.02152065 -0.06141538  0.05536366  0.01151201]
 [-0.05253602  0.03948422 -0.14874639  0.12745512  0.02698888]
 [ 0.03174444 -0.0069775  -0.14866605  0.10300832  0.02369326]
 [ 0.50968476 -0.29557293  0.20675387 -0.30350542 -0.05451889]
 [-0.06123771  0.02312074  0.15027234 -0.09568679 -0.02281405]]
```

```
plt.figure(figsize=(10,7))

Error3 = B - AReduced3
dfError3 = pd.DataFrame(Error3, columns=["Matemáticas", "Ciencias", "Español", "Historia", "EdFísica"])
dfError3.boxplot()
```

```
<Axes: >
<Figure size 1000x700 with 1 Axes>
```

```
print("Errores promedio por columna")
Error3.mean(axis=0)
```

Errores promedio por columna

```
array([-0.00022094,  0.00044883, -0.00462162,  0.00355172,  0.00078359])
```

```
# Por lo que la separacion en 3 dimensiones resulta mas clara que la de 2, ya que si se agrega una
# dimension extra
# se mantiene una mayor cantidad de informacion con los datos originales y tambien se puede
# observar a partir de la
# construccion de los errores los cuales se observan mas pequeños
```